

Navigation Assistance Via Haptic Technology for Blind or Low-Vision Users: A Scoping Review

Proceedings of the Human Factors and Ergonomics Society Annual Meeting 2025, Vol. 69(1) 1–6
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DOI: 10.1177/10711813251360706
journals.sagepub.com/home/pro



Shirley Tang¹  and Gaojian Huang¹ 

Abstract

With estimates of at least 3.4 million Americans aged 40 years and older facing challenges due to vision impairment, advancements in haptic technology have been explored as an effective alternative approach to provide support in aspects such as accessible navigation. A scoping review was conducted to investigate recent literature on haptic technology used to guide blind or low-vision users to a destination, focusing on 28 articles that were identified using the PRISMA method. Results revealed a focus on the methods of haptic input, specifically the benefits and gaps in hands-free and hand-held devices. Several studies utilized the advantages of employing a multimodal feedback approach with audio to better convey navigation information, while explorations of the performance of different types of haptic feedback were also discussed. Overall, this study provides insights into recent advancements in haptic design, which can assist researchers in improving developments toward inclusive navigation technology for blind or low-vision individuals.

Keywords

haptic technology, navigation assistance, blind and low-vision users, vibrotactile feedback, assistive technology, systematic review

Introduction

Individuals with vision impairments face numerous challenges across various facets of life, particularly in an increasingly visually oriented society. According to the Centers for Disease Control and Prevention (CDC), over 3.4 million Americans aged 40 years and older are blind or visually impaired, a number that is expected to double by 2030 (CDC, 2024). Blind or low vision (BLV) individuals are restricted in their ability to live independently in modern-day society, facing hardships in areas such as reading labels, recognizing faces, and performing walking-related activities (Welp et al., 2016). To safely navigate unknown environments, BLV individuals often use traditional mobility aids such as white canes, guide dogs, or other assistive technologies (Williams et al., 2014).

However, mobility tools often have shortcomings in the depth of information communicated as well as in practicality. Current mobility tools can only communicate limited information and may not always be economically feasible. For example, a technique for using white canes involves gliding the cane along a wall to walk straight, which provides a limited range of reach (Williams et al., 2014). With guide dogs, responsibilities such as caring for the dog and financial burdens deter individuals from using one (Varghese et al., 2015). Additionally, these tools lack the ability to provide dynamic

turn-by-turn directions similar to a global positioning system (GPS). Given these limitations, other approaches have been explored to augment deficits by stimulating other senses, such as hearing and touch. Over the past few years, there have been significant efforts to develop solutions for conveying spatial and navigation information to both BLV and sighted individuals using non-visual feedback. Specifically, haptic and tactile feedback have been found to be useful and feasible, especially when an individual's sense of hearing is occupied in noisy environments (Kammoun et al., 2012). Accordingly, many novel solutions, such as vibrotactile wearables and advanced haptic mobility devices, have been developed to enhance aspects of navigation, from spatial perception to helping establish the user's cognitive maps. This presents an opportunity to systematically review solutions that actively guide users to their target destinations using turn-by-turn guidance.

¹Department of Industrial and Systems Engineering, San Jose State University, CA, USA

Corresponding Author:

Gaojian Huang, Department of Industrial and Systems Engineering, San Jose State University, One Washington Square, San Jose, CA 95192, USA.
Email: gaojian.huang@sjsu.edu

While some reviews have investigated accessible navigation assistants or mobile apps, no scoping review has been conducted specifically on haptic technology used to provide turn-by-turn guidance to BLV individuals. Thus, this scoping review aims to synthesize the state-of-the-art literature and findings regarding recent advancements in haptic feedback methods that help blind or low-vision users navigate independently.

Method

This scoping review was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method (Moher et al., 2009) to identify relevant research articles. The search was performed in October 2024, with an initial 960 records identified from 6 databases: Compendex (sourcing from IEEE and ACM) and Inspec via Engineering Village, PubMed, PsycINFO, Scopus, and Web of Science. The search syntax consisted of keywords related to haptics (e.g., “haptic feedback” or “tactile”), blind/low-vision (e.g., “blind” or “visual impairment”), and navigation (e.g., “navigat*” or “wayfinding”) and was limited to peer-reviewed academic journal articles and conference papers written in English. To assess the latest emerging technology, only papers published between 2019 and 2024 were evaluated.

From the initial records, 485 duplicates were removed. After reviewing the titles and abstracts, 393 records were excluded, and an additional 54 records were excluded after full-text screening. Records were selected based on the following inclusion criteria: the technology (1) was tested using a controlled methodology, (2) provided active directional walking guidance to a target point, and (3) evaluated haptic technology for the blind/low-vision population. A total of 28 articles were identified and included in the final analysis. This process is illustrated in Figure 1.

Results

Based on the 28 articles, three main themes emerged regarding haptic technology aiding BLV navigation: haptic input methods, multimodal feedback, and haptic feedback types. For theme 1: methods of haptic input, findings examined hands-free and hand-held approaches. The multimodal feedback section (theme 2) examined studies that utilized audio to complement the haptic experience, and the third theme of haptic feedback types investigated how studies employed vibrotactile, kinesthetic, and other forms of tactile feedback to deliver navigation-related information.

Haptic Input Methods

Among the studies, there are two main categories of input methods through which haptic feedback is delivered to BLV users: Methods that are hands-free (16 papers) and methods that are hand-held (12 papers).

Hands-Free Methods. The hands-free solutions presented were wearable technology targeting different areas of the body. Previous studies have found that vibration feedback on the ears, hands, wrists, neck, and feet provides more perceivability and is more preferred during walking (Dim & Ren, 2017). For example, Zhang, Yao, et al. (2019) explored the use of haptic gloves to transmit vibration patterns, using the left glove for direction guidance and the right for obstacle detection. Though BLV participants showed a response latency of less than 0.4 s, there were concerns regarding the gloves interfering with everyday tasks. For the two studies targeting the wrist, Lee and In (2023) found in the first study that combining both stationary and moving tactons achieved a 100% navigation information recognition rate compared to those in the stationary and moving conditions. In Zhang, Yao, et al. (2019), experiments with healthy, blindfolded and BLV participants measured path-following behavior using a sliding tactile feedback wristband, in which a decreased displacement on the predefined paths was observed in both virtual and real-world test fields. Three studies delivering vibrations on the soles of the feet revealed that direction recognition among healthy, blindfolded participants was over 93% accurate for four main directions (forward, backward, left, right), but less accurate for composed directions such as forward-left and forward-right (Alzighaibi et al., 2021; Kammoun et al., 2023; Tachiquin et al., 2021). In addition to these haptically sensitive regions, there were two studies that explored headwear as an interface for delivering feedback and six studies that focused on the torso and waist. Ma et al. (2024) identified when testing with healthy participants that the frontal and temporal regions of the head are more sensitive than the top in perceiving vibrotactile stimulation, with lower strengths and shorter vibration durations being perceived as most comfortable. Belts were the most commonly utilized device to employ vibration motors at different points around the waist, with a shared approach of improving the user’s spatial orientation. For example, Díaz-Toro et al. (2021) positioned vibrators around the waist and programmed certain motors to vibrate on the side closest to detected obstacles within a user’s vicinity and tested with healthy, blindfolded and BLV participants. On the other hand, Brandebusemeyer et al. (2021) developed a continuous vibration that signals the direction of magnetic north to the user, providing a fixed reference point while navigating.

Hand-Held Methods. The hand-held solutions consisted of integrating haptics into tools such as canes, handles, walkers, and smaller portable devices. Several studies introduced enhancements to or imitations of existing mobility aids, suggesting a focus on incorporating devices into the current BLV experience. Four studies used the standard white cane as a base device and added vibrators along the natural grip of the cane to vibrate the thumb tip or back of hand, providing navigation cues and guidance towards the correct direction (Agrawal et al., 2022; Chaudary et al., 2021; Li, Munoz,

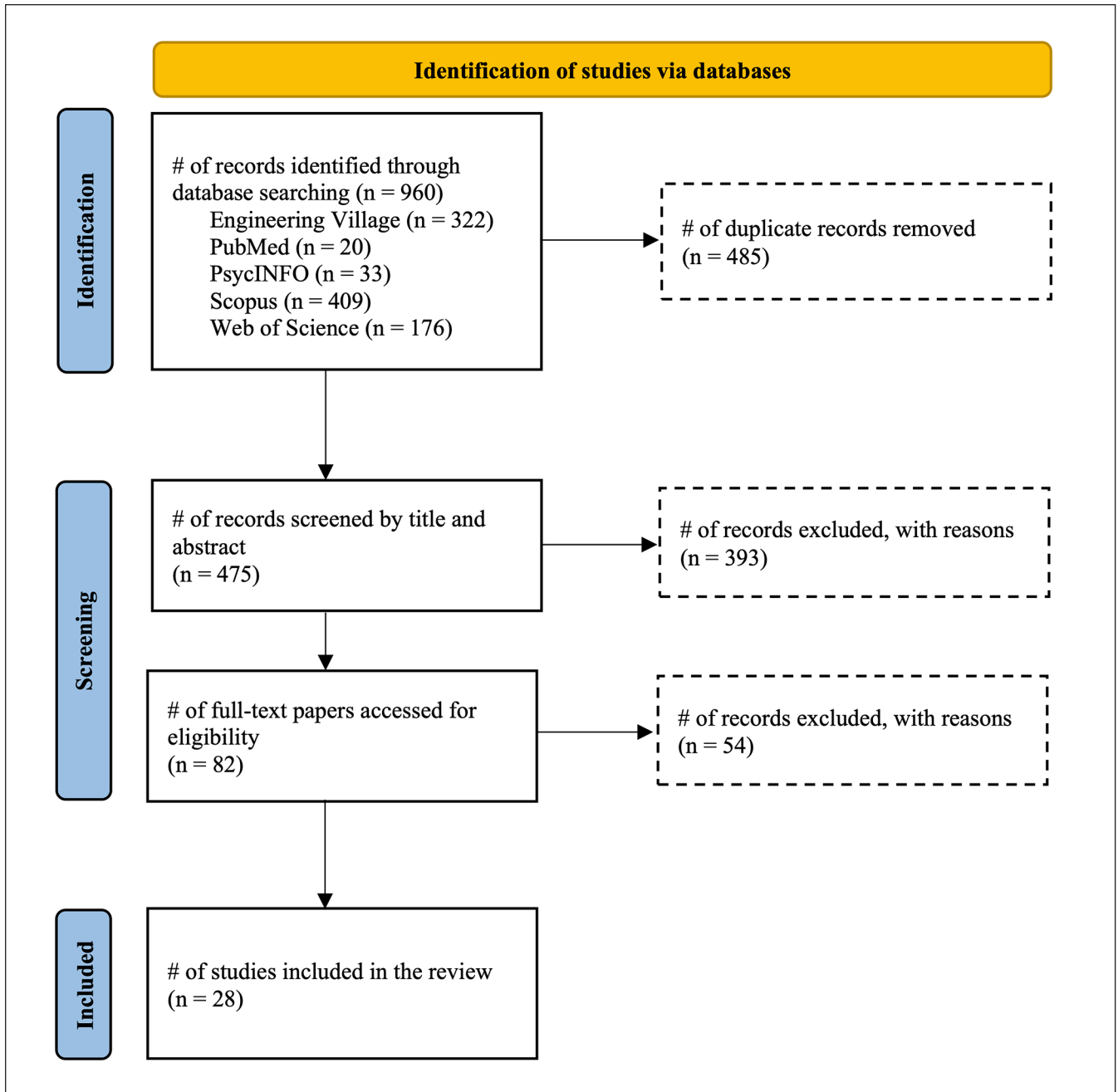


Figure 1. PRISMA flow chart of the process used to select studies.

et al., 2019; Slade et al., 2021). Another approach appealing to existing mobility aids was demonstrated by Shin et al. (2024), where motors were placed into a standalone hand harness to emulate the tactile experience of being led by a guide dog. While the tactile feedback was recognizable, testing on three BLV users with guide dog experience revealed that the force exerted was insufficient for immediate conveyance of actions, suggesting a gap from the actual experience. Additionally, adding vibrators on the handle of a suitcase-shaped base was a proposed solution in two studies: a design

inspired by its ability to move alongside a BLV user like a guide dog or sighted guide. The systems both utilized a LiDAR sensor to enable path planning and to adapt the user's path dynamically to prevent collisions (Guerreiro et al., 2019; Kayukawa et al., 2020). Lastly, studies from Aizawa et al. (2021) with healthy, blindfolded participants and Liu et al. (2021) with BLV participants utilized lighter, smaller portable devices that provided tactile feedback to the thumb. Tests in these studies found that the correct answer rate for identifying directions was always more than 95% regardless

of the hand used to grab the device, and that target directions could be reached with a mean deviation of approximately 3.03° , respectively.

Multimodal Feedback

Along with haptic feedback, 15 studies utilized a multimodal approach incorporating audio feedback to improve information transmission to users, and included evaluations comparing the effectiveness of multimodal and unimodal approaches. Among these studies, categories of audio feedback included verbal instructions for navigation cues (e.g., “turn X degrees to the right”), verbal warnings about the presence of obstacles (e.g., “there’s a man ahead”), verbal descriptions of the environment (e.g., “there’s a kitchen on your right”), and sounds for purposes of immersion (e.g., dog breathing and footsteps to replicate guide dogs; Guerreiro et al., 2019; Li, Muñoz, et al., 2019; Shin et al., 2024; Slade et al., 2021). Of these studies, nine were primarily conducted in controlled indoor laboratory environments (e.g., labs, rooms, or empty spaces with equipment set up), four were conducted in controlled indoor environments such as within existing building structures (e.g., corridors), and four were conducted in structured outdoor environments such as campus grounds and parks (two of which included testing in both indoor and outdoor environments).

Four studies included testing the effectiveness of multimodal feedback. In Zhang, Yao, et al. (2019), researchers conducted tests in a building corridor comparing a pure audio experience with a combined audio-haptics experience using verbal navigation instructions and haptic gloves. Results from testing on four BLV participants revealed that all participants believed the haptic instructions were less likely to cause hesitation compared to solely audio instructions, and that most participants had a better experience with micro-guidance (i.e., precise and continuous directional guidance) in the multimodal approach. Furthermore, a study evaluating the performance of a haptic cane with and without verbal instructions, tested in an indoor 12 by 17 ft room, found that six healthy, blindfolded participants performed better (in terms of navigation time, task completion ratings, and obstacles avoided) when using the cane without the verbal instructions. However, five of six stated they preferred the cane with the verbal instructions (Agrawal et al., 2022).

Haptic Feedback Types

All studies in this review used either vibrotactile, kinesthetic, or other tactile feedback modalities to convey navigation-related information.

Vibrotactile Feedback. Twenty-one studies used vibrotactile feedback to convey various types of information (e.g., navigation instructions, directional cues, obstacle detection alerts). Regarding navigation instructions, six studies using hands-free

technology activated vibrators in specific locations to convey spatial directions. For example, Alzighaibi et al. (2021) defined forward, backward, right, and left directions by vibrating single actuators on the foot sole near the ball of the foot, the heel, and the right and left sides respectively. Alternatively, Guerreiro et al. (2019) used a set of distinct patterns to convey navigation instructions, tested with blind participants: left and right turns were represented by four consecutive vibrations with a 150ms duration and interval, slight right and left turns by two consecutive vibrations of 150ms duration and interval, and going straight by a single vibration of 400ms duration. Two studies delivering vibrations via canes utilized the presence and absence of vibrations to indicate whether the user was facing the right direction, in which vibrations would be felt until the user adjusted themselves to face the correct direction (Chaudary et al., 2023; Li, Munoz, et al., 2019). To aid in obstacle detection along a path, manipulating parameters such as vibration frequency and intensity was used in four studies to indicate proximity to an obstacle and corresponding urgency level, where increasing frequency and intensity levels were associated with increasing proximity to the obstacle and risk of collision. Specifically, in Kayukawa et al. (2020), the system to alert users of collisions defined low urgency warnings by vibrations with a 0.5s pulse duration and 0.5s inter-pulse interval, while high urgency emergency signals were continuous vibrations. Feedback from 14 blind participants in the study showed that participants felt more confident while walking and looking out for alerts, with one participant mentioning they could even walk faster when there were no alerts.

Kinesthetic Feedback. Use of force-feedback was investigated in three studies, demonstrating efficacy in guiding users via kinesthetic feedback. For example, grounded kinesthetic feedback facilitated by a motorized omni wheel placed at the end of a cane was employed by Slade et al. (2021) to steer the user left and right via applied ground-reaction torques. It was found that this method of feedback provided accuracy and efficiency in steering 12 healthy, blindfolded participants toward the correct heading angle, even reaching angles in less time than with vibrotactile feedback. Surichakui et al. (2022) utilized a stretch feedback mechanism that displaced a pin to tension the palmar skin in four cardinal directions, producing unidirectional sensations with an average identification accuracy of 82.5%.

Tactile Feedback. Other types of tactile feedback observed include stimulating the skin through squeezing sensations on the arm, feeling a rotatable needle with the finger, or a sliding tactile sensor on the wrist (Barontini et al., 2021; Liu et al., 2021; Zhang, et al., 2019) to guide users toward the correct direction. For example, Barontini et al. (2021) used normal and tangential force cues to provide indoor navigational guidance via a wearable device on the user’s arm, which was designed to mimic the sensation of being guided by a person’s hand. The normal force was generated by spinning motors in opposite directions to

tighten or loosen the armband (e.g., the cue to walk straight corresponded with two sequential squeezing stimuli), while the tangential force was produced by spinning motors in the same direction (e.g., cues to turn left and right were generated by rotating the motors counterclockwise and clockwise, respectively). Subjective feedback from an experiment with a blind participant who typically relied on a sighted guide indicated increased feelings of safety when walking through a corridor, particularly in relation to obstacle avoidance. In contrast, Liu et al. (2021) introduced a handheld tactile device with a rotatable needle that pointed towards the target direction, with a home marker indicating the user's current position and orientation. Using fingers to feel the deviation of the needle from the marker served as an indicator for the user to adjust their orientation to align the needle and marker. Experiments with 18 BLV participants found that all participants were able to walk along a 60 cm-wide path accurately and smoothly.

Discussion

This scoping review assesses recent literature on the types of haptic technology aimed at providing effective navigation guidance for BLV users. Implications of various body locations for input are included, as well as the effectiveness of using multimodal approaches and specific haptic patterns to convey direction information.

Throughout the review, it was noted that 13 studies tested their solutions with healthy, blindfolded participants with no prior experience using mobility aids. Although this allows for a temporary "blindness simulation," the simulation is still a distorted representation of the actual BLV experience. Therefore, continued research is needed to validate the practicality of these devices with actual BLV users, also highlighting the importance of incorporating these users as part of the research, design, and testing process. Since experiences of low vision extend to populations beyond the BLV community as well, future studies may also choose to investigate the effects of such technology among the elderly or personnel operating in low-visibility environments (e.g., firefighters) in order to evaluate navigation solutions that are scalable across different user groups and domains.

Providing hands-free solutions with wearable haptic technology can be a convenient option for BLV users, who may have their hands occupied with other mobility aids. However, it is important to note that depending on how the device is worn, it can have varying levels of perception to stimuli and may even interfere with everyday functioning if placed on areas such as the hands. Conversely, incorporating haptic feedback into hand-held solutions such as enhancing existing mobility aids like white canes has an advantage of familiarity for BLV users in integrating into their everyday usage habits. Similarly, with hand-held solutions, convenience can be a crucial factor to consider, especially when evaluating the weight of the device and ease of portability, as some studies noted.

The studies evaluating multimodal feedback were primarily tested in controlled indoor and outdoor environments that

may not accurately translate to real-world environments. Because of this, there are opportunities for future research to examine the usability of uni- and multi-modal feedback for navigation in real-world setting such as indoor and outdoor spaces, various weather conditions, and crowded and noisy surroundings. Furthermore, additional research should also investigate the potential effects of environmental noise on the ability to perceive information while navigating in complex real-world situations.

Utilizing different types of feedback and manipulating parameters can be an effective way of conveying information about specific navigation directions or risks of collision. Given that the majority of studies delivered haptic feedback using vibrations, future studies may seek to better explore the effectiveness of conveying guidance and awareness of the environment via other tactile or kinesthetic methods to refine recognition accuracy, navigation time, and user acceptance. Overall, this review maps the current state of haptic technologies for blind or low-vision navigation, and is a foundation for designing more inclusive technologies to enhance one's quality of life and sense of independence.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

Shirley Tang  <https://orcid.org/0000-0002-0984-5208>

Gaojian Huang  <https://orcid.org/0000-0001-8139-7481>

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